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| **Fiscal Year:** | FY 2022 | **Task Last Updated:** | FY 11/30/2022 |
| **PI Name:** | Ostrogorsky, Aleksandar Sc.D. | | |
| **Project Title:** | Diffusion Coefficients of Dopants in Si and Ge Melts | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | MATERIALS SCIENCE--Materials science | | |
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| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | aostrogo@iit.edu | **Fax:** | FY |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | 312-567-5268 |
| **Organization Name:** | Illinois Institute of Technology | | |
| **PI Address 1:** | Mechanical, Materials and Aerospace Engineering Department | | |
| **PI Address 2:** | 10 W 31st St | | |
| **PI Web Page:** | | | |
| **City:** | Chicago | **State:** | IL |
| **Zip Code:** | 60616-3729 | **Congressional District:** | 7 |
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| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA MSFC |
| **Contact Monitor:** | Sansoucie, Michael | **Contact Phone:** | 256.544.5269 |
| **Contact Email:** | michael.p.sansoucie@nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Volz, Martin Ph.D. ( NASA Marshall Space Flight Center ) | | |
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| **Performance Goal Text:** | | | |
| | The compositional homogeneity and electronic properties of single crystal semiconductors depend on the diffusion coefficients of impurities and dopants dissolved in the melt. However, even after several decades of research, the published values of diffusion coefficients can vary significantly. Diffusion is a slow process, and thus affected by minute levels of convection. Thus, it is expected that previous measurements of diffusion have been compromised by natural or buoyancy-driven convection, which is unavoidable on Earth. Microgravity presents an optimal environment to obtain high-fidelity diffusion data. This data is essential for the computer models which are used to optimize industrial crystal growth processes. The key goals of this proposal are as follows:<br>(i) Develop and calibrate a process to measure diffusion coefficients on Earth. | | |

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| **Task Description:** | (ii) Use this process to make accurate measurements of diffusion coefficients in Ge in microgravity.<br><br>(iii) Use this process to measure the diffusion coefficients of impurities in Si melts.<br><br>We propose to conduct on Earth and in microgravity the following experiments:<br><br>(a) Bridgman growth experiments using bundles of small diameter ampules.<br><br>(b) Direct measurements of diffusion coefficients in bundles of capillary tubes. The key goal is to determine precisely the diffusion coefficients of Ga, B, Sb, and Si in germanium.<br><br>Furthermore, on Earth, following the pioneering Burton-Prim-Slichter (BPS) papers [1,2] we propose to conduct (c) Czochralski crystal growth experiments, at high RPM, from small melts, in order to obtain steady laminar symmetric flow (eliminate unsteady natural convection) and use the BPS model and finite element (FE) modeling, to measure the diffusion confidents in Ge and Si melts.<br><br>In preparation for the microgravity experiments, the following ground based experiments will be conducted:<br><br>(i) Direct measurements of diffusion coefficients in spiral or circular capillary tubes in the 5 Tesla axial magnet at Marshall Space Flight Center (MSFC).<br><br>(ii) Direct measurements of diffusion coefficients using bundles of straight capillary tubes in 1 Tesla transverse magnet at Illinois Institute of Technology (IIT).<br><br>(iii) Bridgman growth experiments using a bundles of small diameter ampoules inside the Sample Cartridge Assembly (SCA).<br><br>The microgravity experiments will utilize the Low Gradient Furnace (LGF) furnace in the Materials Science Research Rack (MSRR). The experiments will consist of both “direct” measurements of diffusion using small capillaries as well as directional solidification experiments. The variety of experiment variables will provide a surfeit of relevant diffusion data in alignment with the goals of the MaterialsLab initiative. These include the variety of dopants to be measured, with varying segregation coefficients, the variation in processing parameters such as temperature, and the ability to process up to 7 small capillaries in a single experiment.<br><br>References:<br><br>[1] J.A. Burton, R.C. Prim, W.P. Slichter. The Distribution of Solute in Crystals Grown from the Melt. Part I. Theoretical. Journal of Chemical Physics 21:1987, 1953.<br><br>[2] J.A. Burton, R.C. Prim, W.P. Slichter. The Distribution of Solute in Crystals Grown from the Melt. Part II. Experimental. The Journal of Chemical Physics 21:1991, 1953. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | |
| **Task Progress:** | Several design options for the ampoule with the capillaries were considered and tested. Our initial plan was to develop an ampoule that could be used both at the International Space Station (ISS) and in our laboratory. We considered and tested the setup in both horizontal and vertical orientation. Finite element simulations of diffusion and convection in the capillaries were conducted under terrestrial and microgravity.<br>Our present focus is on using relatively short “finite” capillaries (typically 40 mm long) that appear to be superior to the longer “semi-infinite” capillaries. The short capillaries are particularly suitable for the future flight experiments at the ISS. An ampoule with “finite” capillaries (L=120 mm, OD=12 mm) could fit into the small furnace, Microgravity Research for Versatile Investigations (M.a.R.V.I.n.) which Tec-Masters, Inc. is developing at present. In 2021, the experiments in our laboratory at the Illinios Institute of Technology (IIT) were conducted in a horizontal two-zone clam-shell split furnace. In 2022, we switched to a vertical six-zone tubular furnace which provided a solid axial temperature profile but so far has not produced satisfactory results. Our latest attempt is to conduct diffusion experiments using a Czochralski-type Arthur D. Little (ADL) furnace, which has been modified and committed to this project. A significant progress was made in simulating the transient diffusion and melt convection in capillaries using the finite element code in ANSYS. |
| **Bibliography Type:** | Description: (Last Updated: 05/30/2024) |
| **Abstracts for Journals and Proceedings** | Ostrogorsky AG Volz M, Croell A. "Measurement of diffusion coefficients of dopants in Ge and Si melts." Materials Research Society Fall Meeting, San Diego, California, February 26, 2020.<br>Abstracts. Materials Research Society Fall Meeting, San Diego, California, February 26, 2020. , Feb-2020 |
| **NASA Technical Documents** | Ostrogorsky AG, Volz M, Croell A. "Measurement of Diffusion Coefficients of Dopants in Ge and Si Melts," Huntsville, Alabama: NASA Marshall Space Flight Center, 2020. NASA/No. M20-7971.<br>https://ntrs.nasa.gov/api/citations/20200001746/downloads/20200001746.pdf , Feb-2020 |